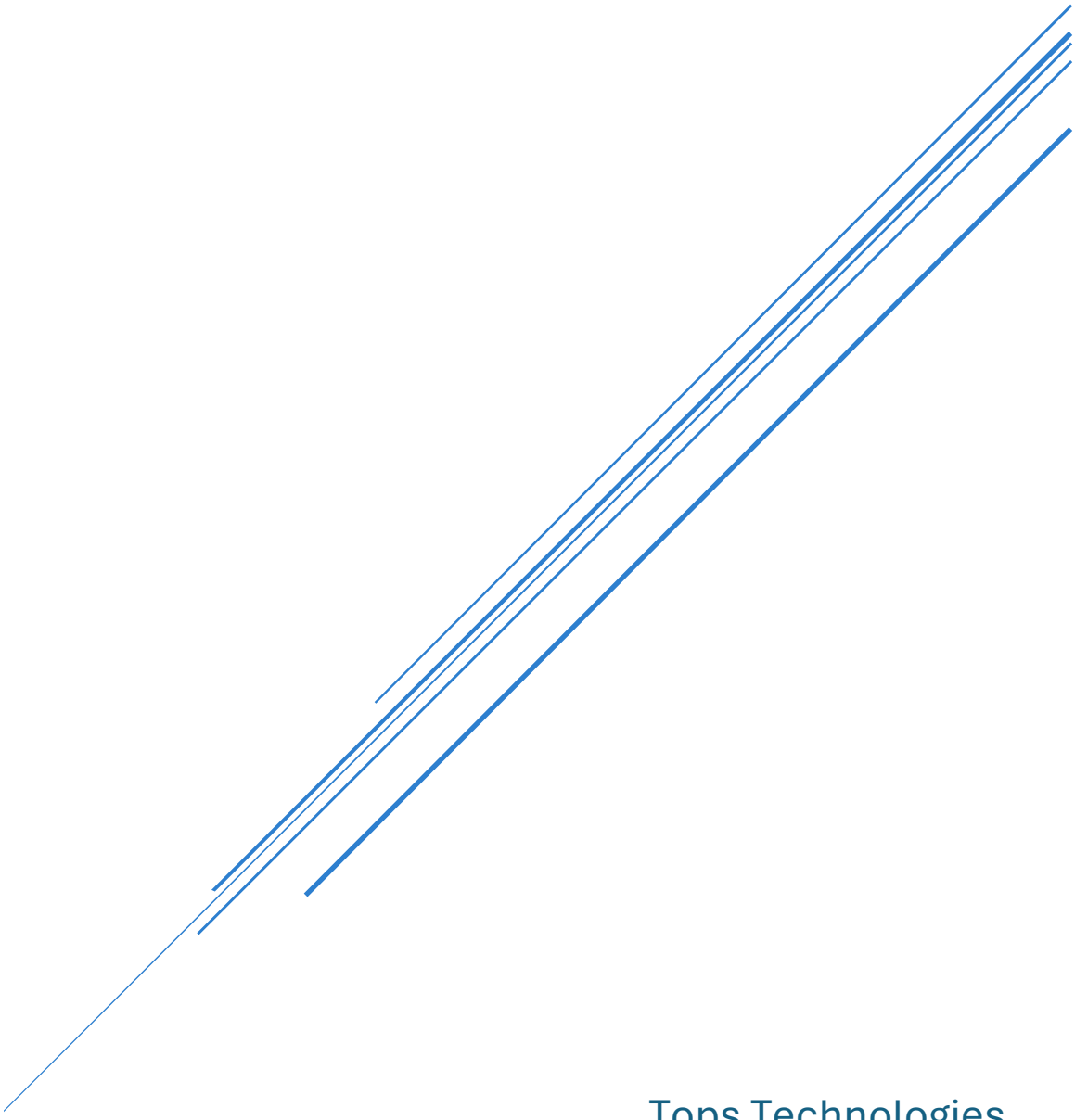
DAY – 2 TCP/IP SUITE

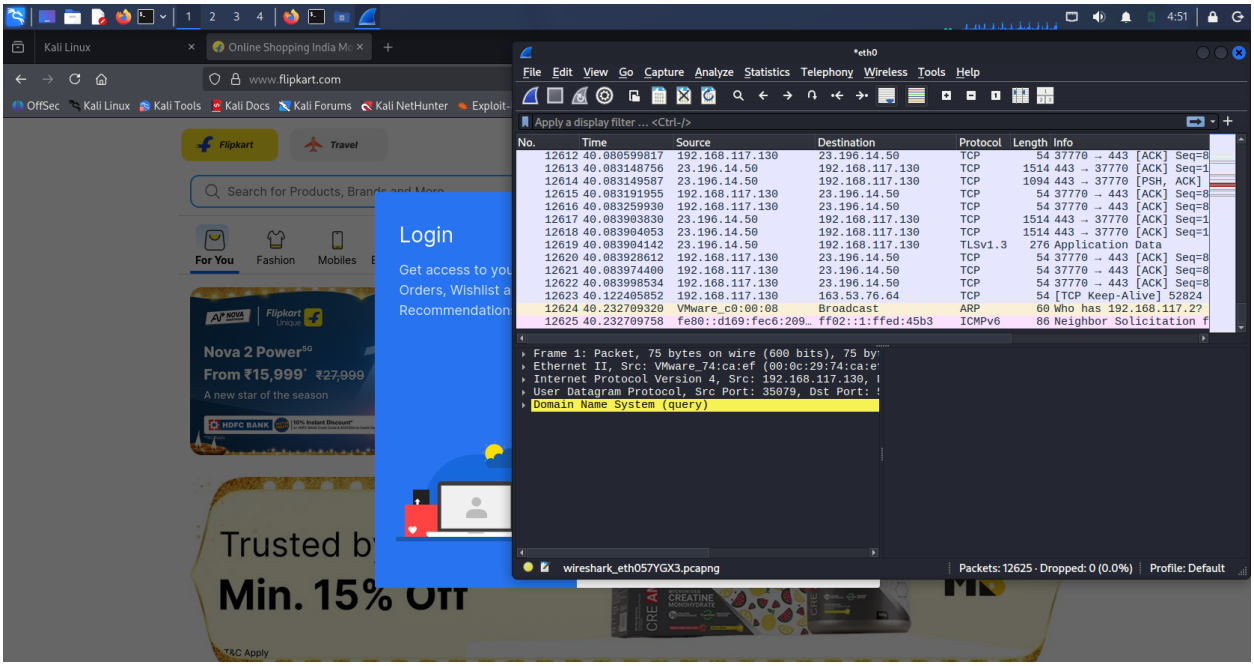
Assignment - 13



Tops Technologies
Cyber Security

1. **Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.**

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Identification of TCP/IP Model Layers Using Wireshark

Layer	Captured Protocol	Evidence from Wireshark
1. Application	DNS / TLSv1.3	DNS query packet details and TLSv1.3 Application Data are visible.
2. Transport	TCP	TCP packets show source and destination ports, such as 37770 → 443.
3. Internet	IPv4	The packet details show IPv4 source and destination addresses, such as 192.168.117.130 and 23.196.14.50.
4. Network Access	Ethernet II	Packet details show Ethernet II and the source MAC address.

2. **Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).**

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Comparison of OSI and TCP/IP Models

OSI Model Layer	TCP/IP Model Layer	Real-World Example
7. Application	Application	Sending a message on WhatsApp.
6. Presentation	Application	Encryption and image formatting when sharing photos on Instagram.
5. Session	Application	Maintaining a communication session during a WhatsApp call.
4. Transport	Transport	TCP helps deliver data reliably when loading a website.
3. Network	Internet	IP addresses route a Zomato order's data to the correct server.
2. Data Link	Network Access	Wi-Fi or Ethernet transfers data between a laptop and router.
1. Physical	Network Access	Wi-Fi radio signals or Ethernet cables carry data as signals.

3. **Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.**
   Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

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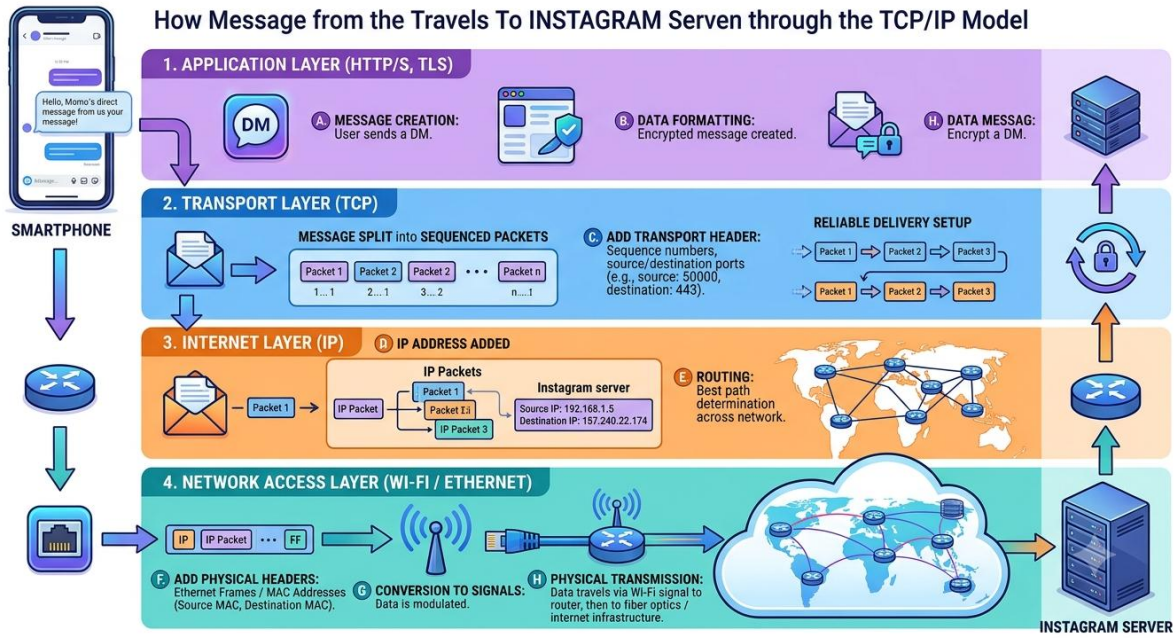
TCP Header Field	Function
Source Port	Identifies the port number of the application sending the data.
Destination Port	Identifies the port number of the application receiving the data.
Sequence Number	Indicates the position of data bytes in the transmission, helping arrange data in the correct order.
Acknowledgment Number	Indicates the next byte expected from the sender, confirming received data.
Header Length (Data Offset)	Specifies the size of the TCP header and where the actual data begins.
Flags (Control Bits)	Control connection operations, such as establishing, maintaining, and closing a connection (SYN, ACK, FIN, and RST).
Window Size	Indicates how much data the receiver can accept before requiring further acknowledgment.
Checksum	Helps detect errors in the TCP header and data during transmission.
Urgent Pointer	Indicates the position of urgent data when the URG flag is set.
Options	Contains optional settings, such as Maximum Segment Size (MSS) and window scaling.
Data (Payload)	Contains the application data being transmitted. It is not part of the TCP header itself.

Conclusion: TCP header fields help identify applications, maintain the correct order of data, detect transmission errors, and provide reliable communication between devices.

4. **Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each**

step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

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When I send a message using Instagram, the data passes through four layers of the TCP/IP model.

1. Application Layer: Instagram prepares the message and communicates with its server.

2. Transport Layer: TCP or UDP carries the data between the device and the server, depending on the connection used.

3. Internet Layer: IP addresses are used to route data packets toward the Instagram server.

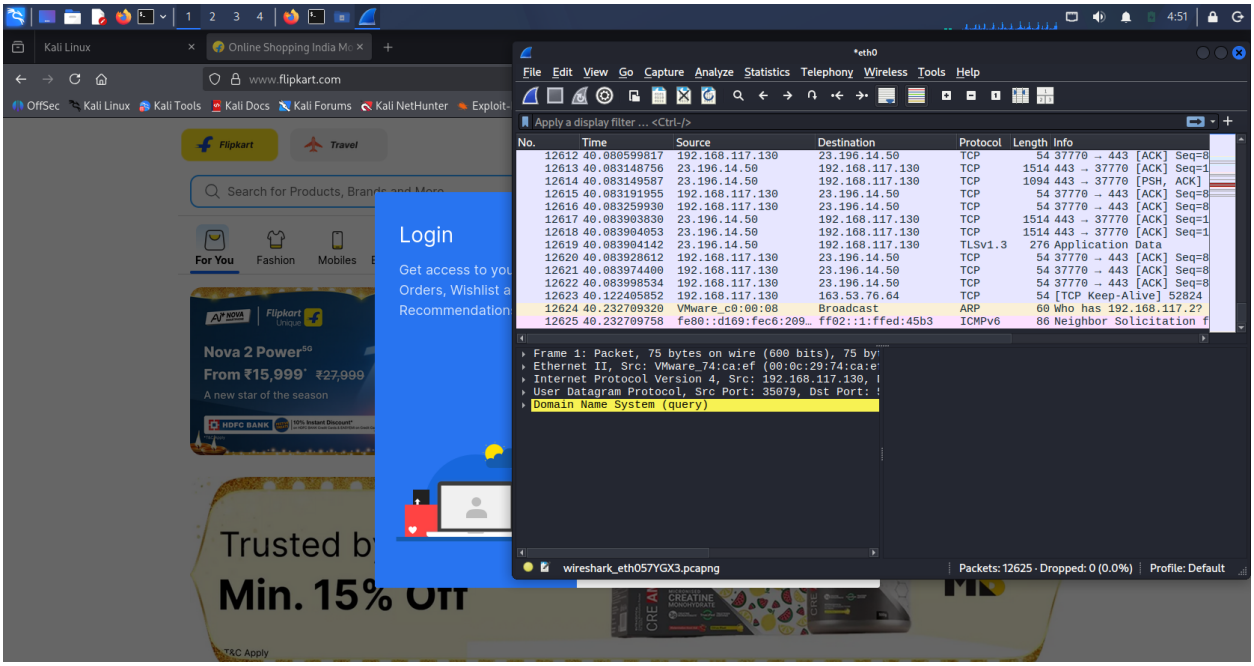
4. Network Access Layer: The data is transmitted through Wi-Fi or Ethernet as signals across the network.

When the data reaches the Instagram server, it is processed and the server sends a response back to the device.

5. **Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like**

sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

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Wireshark Packet Analysis

1. Selected Packet: TCP

2. TCP/IP Layer: Transport Layer

3. How I Identified It: I identified the packet by checking the Protocol column in Wireshark, where TCP is displayed. The Info column shows source and destination ports, such as 37770 → 443.

4. Real-World Action: This packet represents data communication between my device and the Flipkart server while I was browsing the Flipkart homepage.